

Problem 6

Problem 1 Consider a Riemann sum with n rectangles for the function f on the interval $[a, b]$.

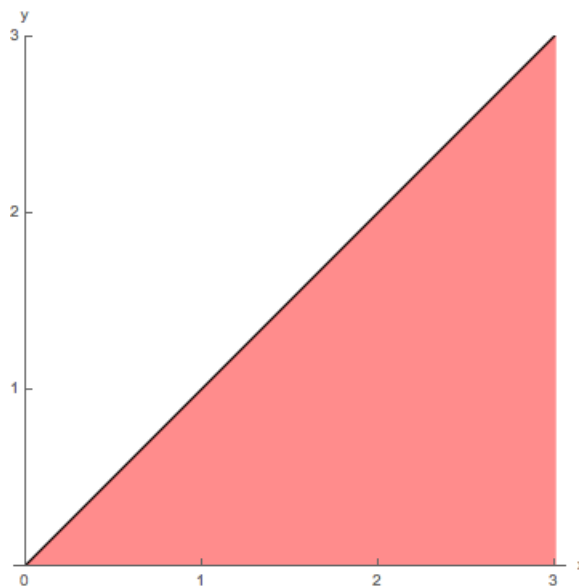
Use geometry to find the limit of Riemann sums as $n \rightarrow \infty$.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = ?$$

(a) $f(x) = x$, $[0, 3]$;

Explanation. $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = \lim_{n \rightarrow \infty} \sum_{k=1}^n x_k^* \Delta x = \frac{9}{2},$

since $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = A$, the area of the shaded region in the figure.



(b) $f(x) = |x|$, $[-3, 3]$;

Explanation. $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = \lim_{n \rightarrow \infty} \sum_{k=1}^n |x_k^*| \Delta x = 9,$

since $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = A$, the area of the shaded region in the figure.

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